Trishit Mukherjee

Mathematically grounded deep learning for inverse imaging and stable reconstruction

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Research Profile

Junior Research Fellow at IISc working at the intersection of inverse problems, computational imaging, operator learning, and optimization theory. Developed expressive neural denoisers and learned regularizers with provable convergence guarantees for iterative image reconstruction with GPU implementations in PyTorch, and CUDA acceleration. Seeking a PhD in inverse problems, computational imaging, theoretically grounded computer vision.

Education

M.Tech. in Artificial Intelligence, Indian Institute of Science, Bengaluru 2023–2025
CGPA: 8.4/10

B.E. in Computer Science and Engineering, Jadavpur University, Kolkata 2019–2023
CGPA: 8.81/10

Research Projects and Publications

Contractive Neural Networks for Stable Image Reconstruction *May 2024–Jun 2025*
Master's Thesis · Prof. Kunal N. Chaudhury, IISc

- **Stabilizing Deep Reconstruction Operators with Contractive Anchoring** [arXiv | code]
  A. Sinha, T. Mukherjee, and K. N. Chaudhury. *European Conference on Computer Vision (ECCV), 2026.*
- **ViSTA: Viscosity Stabilized Plug-and-Play Reconstruction** A. Sinha, T. Mukherjee, and K. N. Chaudhury.
- Formalized peak-and-collapse instability via a local stability index and developed adaptive contractive anchoring to stabilize black-box PnP/RED operators without retraining the pretrained denoiser.
- Designed contractive gradient-step CNN anchors using spectral constraints and learnable splines; validated on deblurring, superresolution, and MRI, sustaining 0.5–1 dB PSNR gains across iterations.

Trainable Kernel Denoisers for Convergent Image Reconstruction *Jan 2026–Present*
Prof. Kunal N. Chaudhury, IISc

- **Trainable Nonexpansive Denoisers for Contractive Image Reconstruction** [arXiv | code]
  A. Sinha, A. Banerjee, T. Mukherjee, and K. N. Chaudhury. *International Conference on Machine Learning (ICML), 2026.*
- Proposed a permutation-based weighted-aggregation denoiser whose global *nonexpansiveness* is guaranteed by construction (permutation group action + a symmetrization protocol), using a compact weight predictor.
- Proved that the induced PnP reconstruction map is *contractive* under mild conditions, yielding global convergence and remaining competitive performance on deblurring and superresolution against state-of-the-art baselines.

`neural_shift_cuda`: Parallel Kernel Aggregation through CUDA acceleration *Jul 2026–Present*
Open-source C++ / CUDA extension · Lead developer

- Fused serial shift aggregation into parallel CUDA kernels with full autograd, a PyTorch CPU fallback, and reference-equivalence tests, preserving the original models and operators.
- Achieved 2.23×/1.31× full-step speedups for improved symmetric/asymmetric kernel denoisers and 1.61×/1.33× for the original NECTR implementations; supports Pascal–Blackwell GPUs.

Geosocial Road-Accident Data Analysis *Aug 2022–May 2023*
Research Project · Prof. Sarbani Roy, Jadavpur University

- **Location Metadata Extraction from Geosocial Data of Road Accidents using Deep Learning Models**
  T. Mukherjee, S. Sinhahajari, D. Mukherjee, H. Mallick, A. I. Middya and S. Roy. *Evolving Systems*, 16(1), 2025.
- Built an API-based news pipeline with LSTM/Word2Vec relevance classification (96% accuracy, 0.94 F1) and BERT location extraction (0.673 exact match).

Work Experience and Teaching

Junior Research Fellow · LISA, EE, IISc *Aug 2025–Present*
Supervisor: Prof. Kunal N. Chaudhury · Bengaluru, India
Developing stochastic denoisers and graph/kernel-Laplacian regularizers using learned edge weights, conjugate gradient solves and implicit differentiation; expansive operators for provably convergent reconstruction.

Teaching Assistant · Linear Algebra and Applications, IISc *Aug–Nov 2024*
Instructor: Prof. Kunal N. Chaudhury · Bengaluru, India
Ran tutorials and helped develop assignments for undergraduate and graduate students.

Software Engineering Intern · Innovation Junction LLC *Jun–Aug 2021*
Remote, USA
Built AWS data pipelines for CSV operations, SQL/NoSQL ingestion into EC2, and Power BI reporting.